

Chapter 02: First-Order Differential Equations & Modeling

Concordia University · Department of Mechanical, Industrial & Aerospace Engineering

Course: Applied Ordinary Differential Equations (ENGR 213) | **Textbook:** Official Course Textbook (7th Ed., Chapter 2)

1. Executive Summary & First-Principles Philosophy

Chapter 2 forms the analytical and practical backbone of first-order differential equations. In this chapter, we master the 5 core analytical solution techniques:

1. **Separation of Variables** ($g(y)dy = f(x)dx$)
 2. **Integrating Factors for Linear Equations** ($y' + P(x)y = Q(x)$)
 3. **Exact Equations & Multipliers** ($Mdx + Ndy = 0$)
 4. **Substitutions** (Homogeneous, Bernoulli, Linear Composition)
 5. **Applied Physical Modeling** (Newton's cooling, brine tanks, logistic population dynamics)
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2. Core Mechanics & Mathematical Engine

A. Separation of Variables

If an ODE can be factored as $\frac{dy}{dx} = g(x)h(y)$, divide by $h(y)$ (noting any zero-division singular solutions where $h(y) = 0$) and integrate both sides:

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C$$

B. First-Order Linear ODEs & The Integrating Factor

Standard linear form requires the leading coefficient of $\frac{dy}{dx}$ to be strictly 1:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

Multiply every term by the integrating factor $\mu(x) = e^{\int P(x)dx}$:

$$e^{\int P(x)dx} \frac{dy}{dx} + P(x)e^{\int P(x)dx} y = Q(x)e^{\int P(x)dx}$$

Recognize the left side as the product rule: $\frac{d}{dx}[\mu(x)y] = \mu(x)Q(x)$. Integrate and divide:

$$y(x) = \frac{1}{\mu(x)} \left[\int \mu(x)Q(x)dx + C \right]$$

C. Exact Equations & Integrating Multipliers

An equation $M(x, y)dx + N(x, y)dy = 0$ is **exact** if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Then there exists a potential function $F(x, y)$ such that $\frac{\partial F}{\partial x} = M$ and $\frac{\partial F}{\partial y} = N$. The general solution is $F(x, y) = C$.

If not exact, calculate:

1. If $\frac{M_y - N_x}{N}$ depends only on x , then $\mu(x) = \exp\left(\int \frac{M_y - N_x}{N} dx\right)$.
2. If $\frac{N_x - M_y}{M}$ depends only on y , then $\mu(y) = \exp\left(\int \frac{N_x - M_y}{M} dy\right)$.

D. Substitution Archetypes

1. **Homogeneous Equations:** If $M(tx, ty) = t^n M(x, y)$ and $N(tx, ty) = t^n N(x, y)$, substitute $y = ux \implies dy = udx + xdu$.
2. **Bernoulli Equation:**

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Divide by y^n : $y^{-n} \frac{dy}{dx} + P(x)y^{1-n} = Q(x)$. Substitute $w = y^{1-n}$, which transforms into linear:

$$\frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$$

3. **Linear Composition:** $\frac{dy}{dx} = f(Ax + By + C) \implies u = Ax + By + C \implies \frac{du}{dx} = A + B \frac{dy}{dx}$.
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3. Curriculum-Grounded Visual Reference & Textbook Figure Breakdowns

Visual 3.1: Direction Fields & Solution Curves (Textbook Fig. 2.1.2)

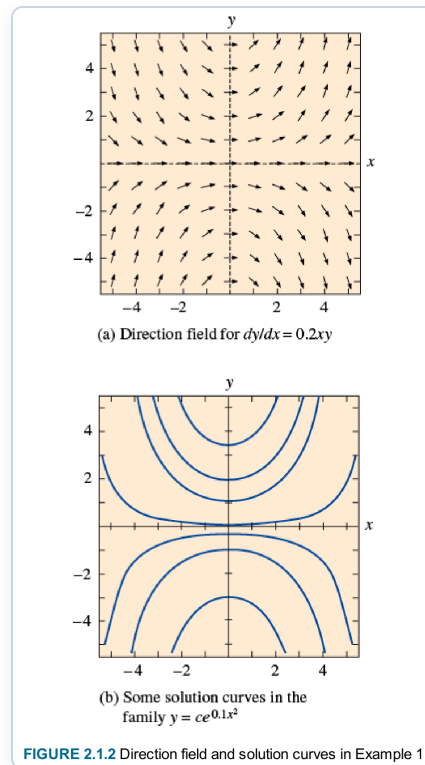


Figure 2.1: Direction field and family of integral solution curves for $\frac{dy}{dx} = 0.2xy$ — extracted directly from Official Textbook (7th Ed., Chapter 2, Fig. 2.1.2).

In-Depth Pedagogical Breakdown:

1. Lineal Elements & Tangent Grids (Fig. 2.1.2a):

- At every Cartesian coordinate (x, y) , the differential equation $\frac{dy}{dx} = 0.2xy$ prescribes a unique numerical slope.
- Coordinate Symmetries:**
 - On the axes ($x = 0$ or $y = 0$), the slope is $0.2(0) = 0$. Hence, every lineal element along the x -axis and y -axis is strictly horizontal.
 - In Quadrant I ($x > 0, y > 0$) and Quadrant III ($x < 0, y < 0$), the product $xy > 0$, so the slopes are strictly positive (pointing upward to the right).
 - In Quadrant II ($x < 0, y > 0$) and Quadrant IV ($x > 0, y < 0$), the product $xy < 0$, so the slopes are strictly negative (pointing downward to the right).

2. Integral Solution Curves (Fig. 2.1.2b):

- Solving $\frac{dy}{dx} = 0.2xy$ by separation of variables:

$$\frac{dy}{y} = 0.2x dx \implies \ln |y| = 0.1x^2 + C_1 \implies y(x) = ce^{0.1x^2}$$

- Any trajectory dropped into the direction field must remain everywhere tangent to the lineal elements.
- For $c > 0$, the curves form upward-opening symmetric exponential wells.

- For $c < 0$, the curves form downward-opening symmetric profiles.
- The line $y \equiv 0$ ($c = 0$) serves as an exact equilibrium barrier that solution curves never cross due to Picard-Lindelöf uniqueness.

Visual 3.2: Physical Mixing Tank Architecture & Salt Accumulation (Textbook Fig. 1.3.3 & Fig. 2.7.4)

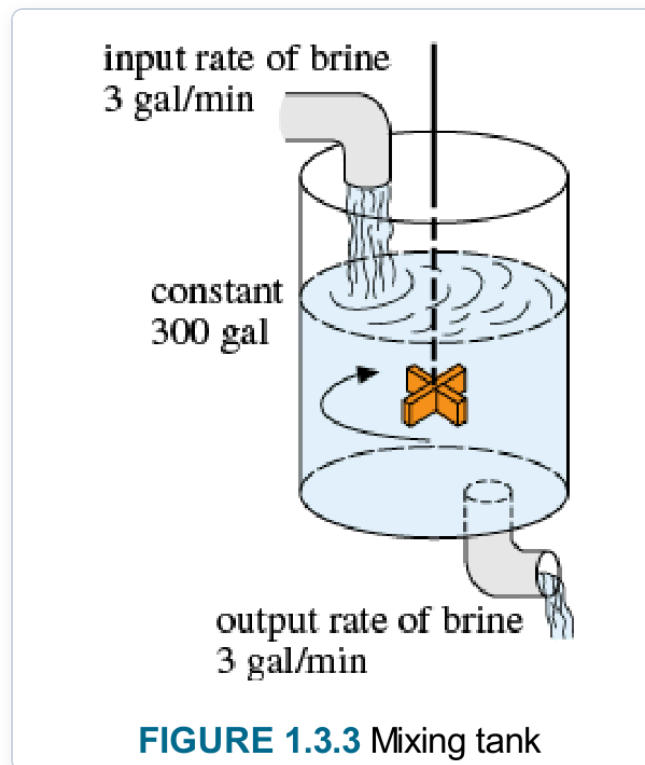


Figure 2.2: Mixing tank schematic showing fluid inflow, impeller agitation, and bottom drainage — from Textbook 7th Ed. (Fig. 1.3.3).

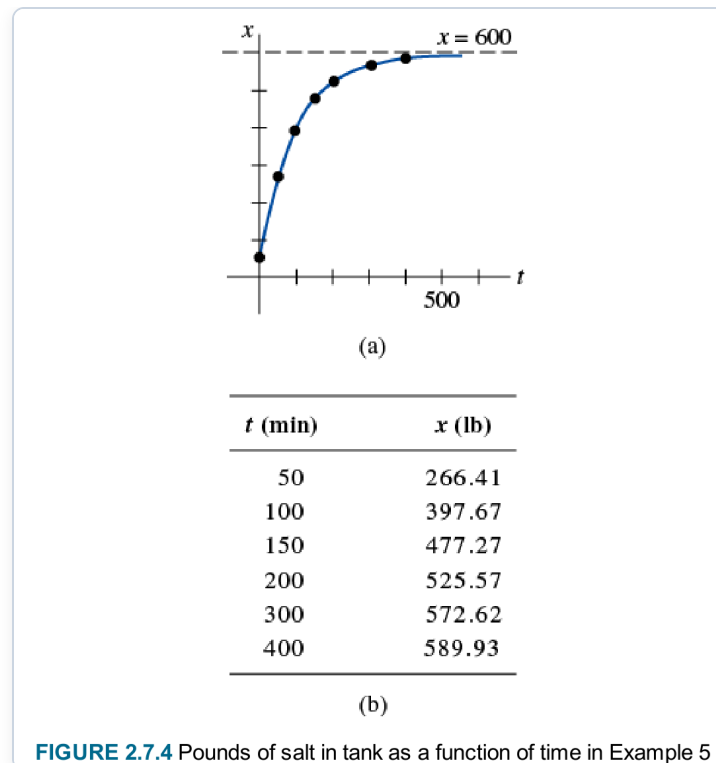


Figure 2.3: Salt accumulation curve $x(t)$ approaching horizontal asymptote $x = 600$ lb — from Textbook 7th Ed. (Fig. 2.7.4).

In-Depth Pedagogical Breakdown:

1. Fluid Dynamics & Mass Conservation (Fig. 1.3.3):

- **Inflow Stream:** Brine enters the top pipe at volumetric flow rate $r_{\text{in}} = 3$ gal/min with dissolved salt concentration $c_{\text{in}} = 2$ lb/gal, delivering salt input $R_{\text{in}} = r_{\text{in}} \cdot c_{\text{in}} = 6$ lb/min.
- **Internal Tank Agitation:** An impeller continuously mixes the fluid to ensure uniform spatial concentration throughout the 300 gallon volume.
- **Drainage Stream:** Well-stirred brine exits through the bottom pipe at $r_{\text{out}} = 3$ gal/min, carrying salt out at instantaneous rate $R_{\text{out}} = r_{\text{out}} \cdot \frac{A(t)}{V(t)} = \frac{3}{300}A(t) = \frac{1}{100}A(t)$ lb/min.

2. Dynamic Trajectory to Equilibrium (Fig. 2.7.4):

- The governing linear ODE $\frac{dA}{dt} + \frac{1}{100}A = 6$ yields the analytical solution $A(t) = 600 - (600 - A_0)e^{-t/100}$.
- **Transient Phase:** As shown in the graph and table, the salt content rapidly climbs from $A(0) = 50$ lb to 266.4 lb at $t = 50$ min, and 525.6 lb at $t = 200$ min.
- **Steady-State Saturation:** As $t \rightarrow \infty$, the exponential transient dies out ($e^{-t/100} \rightarrow 0$), and the salt content asymptotically approaches $A_{\text{limit}} = 600$ lb ($V \times c_{\text{in}} = 300 \text{ gal} \times 2 \text{ lb/gal} = 600 \text{ lb}$).

Visual 3.3: One-Dimensional Phase Line & Autonomous Trajectories (Textbook Fig. 2.1.6)

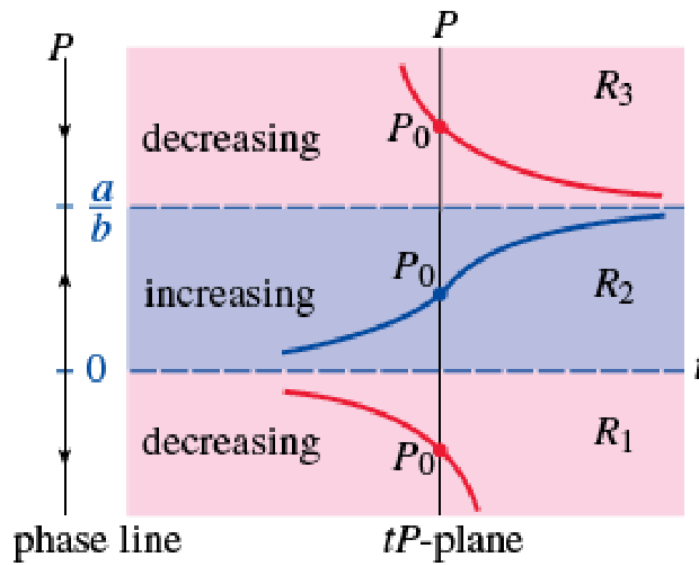


FIGURE 2.1.6 Phase portrait and solution curves in each of the three subregions in Example 4

Figure 2.4: Autonomous phase line mapped to solution trajectories in the tP -plane across regions R_1, R_2, R_3 — from Textbook 7th Ed. (Fig. 2.1.6).

In-Depth Pedagogical Breakdown:

1. Equilibrium Lines:

- The horizontal dashed lines $P = 0$ and $P = a/b$ represent equilibrium solutions where $\frac{dP}{dt} = 0$.
- These equilibrium lines partition the phase plane into three distinct invariant regions: R_1 ($P < 0$), R_2 ($0 < P < a/b$), and R_3 ($P > a/b$).

2. Flow Direction & Stability:

- In region R_2 , $\frac{dP}{dt} > 0$ (indicated by upward arrow on the phase line). Solutions starting at $P_0 \in (0, a/b)$ increase monotonically toward the carrying capacity a/b , exhibiting an S-shaped logistic curve with an inflection point.
- In region R_3 , $\frac{dP}{dt} < 0$ (indicated by downward arrow). Solutions starting above carrying capacity decrease asymptotically toward a/b .
- Hence, $P = a/b$ is an **asymptotically stable sink (attractor)**, while $P = 0$ is an **unstable source (repeller)**.

4. Fully Worked Exam Archetype: The Draining Mixture Tank

Problem: A tank initially contains $V_0 = 300$ gal of brine in which $A_0 = 50$ lb of salt is dissolved. Pure water flows in at $r_{\text{in}} = 3$ gal/min, and the thoroughly mixed solution pumps out at $r_{\text{out}} = 2$ gal/min. Find the salt content $A(t)$ at any time t .

Step-by-Step Solution:

- **Step 1: Governing Mass Balance:**

$$\frac{dA}{dt} = R_{\text{in}} - R_{\text{out}} = (c_{\text{in}}r_{\text{in}}) - (c_{\text{out}}(t)r_{\text{out}})$$

Since pure water enters, $c_{\text{in}} = 0 \implies R_{\text{in}} = 0$.

- **Step 2: Volume Formulation:**

$$V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t = 300 + (3 - 2)t = 300 + t \text{ gal}$$

Therefore, concentration at time t is $c_{\text{out}}(t) = \frac{A(t)}{300+t}$.

- **Step 3: Setup First-Order Linear ODE:**

$$\frac{dA}{dt} = -\frac{2A}{300+t} \implies \frac{dA}{dt} + \frac{2}{300+t}A = 0$$

- **Step 4: Separation / Integrating Factor:**

$$\frac{dA}{A} = -\frac{2}{300+t}dt \implies \ln|A| = -2\ln(300+t) + C_1$$

$$A(t) = \frac{C}{(300+t)^2}$$

- **Step 5: Apply Initial Condition:**

$$A(0) = 50 \implies 50 = \frac{C}{300^2} \implies C = 50 \times 90000 = 4,500,000$$

$$A(t) = \frac{4,500,000}{(300+t)^2} \text{ lb}$$

5. Rapid Exam Traps & Red Flags

- **⚠ Trap 1: Dropping $+C$ Before Exponentiating:** In separable equations, write $+C$ immediately after integrating. $e^{\ln|y|} = e^{kt+C} = C_0e^{kt}$. Never add $+C$ at the very end as an afterthought!
- **⚠ Trap 2: Variable Volume Tank Denominators:** When inflow rate $\neq$ outflow rate, volume is dynamic: $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$. Forgetting this produces an incorrect constant denominator.